EXPERIMENT -4: CODE

```
const int LM35_PIN = A0;
const int PWM_PIN = 9;      // PWM output
const int BUZZER_PIN = 7;   // BC547 base through 1k resistor

const float TEMP_LIMIT = 30.0;

void setup() {
  pinMode(PWM_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  // Read LM35 sensor
  int sensorValue = analogRead(LM35_PIN);

  // Convert analog reading to temperature
  float voltage = sensorValue * (5.0 / 1023.0);
  float temperatureC = voltage * 100.0;

  // Map temperature from 0-50 C to PWM 0-255
  int pwmValue = constrain(
    map((int)(temperatureC * 10), 0, 500, 0, 255),
    0, 255
  );

  // Output PWM signal
  analogWrite(PWM_PIN, pwmValue);
```

```
  // High-temperature alert
  digitalWrite(BUZZER_PIN,
               temperatureC >= TEMP_LIMIT ? HIGH : LOW);

  // Display temperature and PWM
  Serial.print("Temperature: ");
  Serial.print(temperatureC, 1);
  Serial.print(" C | PWM: ");
  Serial.println(pwmValue);

  delay(1000);
}
```